

## Credit Card Validation Service

### Summary of Requirements

You will create a Card Validation Service using Microsoft Visual Studio .NET. The ideal tools will allow users to validate credit card numbers interactively via an XML web service and ASP.NET.

The objectives of the project are focused on driving progress and improvement in the following key areas:

- Utilize Internet Information Services (IIS) authentication to manage access to your XML web services.
- Create client applications that utilize credentials to authenticate with secure XML Web services.
- Utilize ASP.NET authorization controls to limit access to XML Web service resources.

The finished application shall include the following:

- XML Web Service Method
- Client Application
- Authentication Process
- Business function: Credit Card Validation Process (features):
  - Ensure that the credit card number has been issued by a valid provider
  - Ensure that the credit card has not expired
  - Ensure that the credit card has not been canceled by the appropriate card provider
  - Ensure that the credit card holder has sufficient credit to enter a transaction

#### 1. Create XML Web service for Credit Card Validation Process

**Objective:** Please create validation methods for the following credit card types: American Express (AMEX), VISA, and MasterCard.

##### Method

|                    |                                                 |
|--------------------|-------------------------------------------------|
| ValidateAMEX       | Validate American Express credit or charge card |
| ValidateVisa       | Validate VISA credit card                       |
| ValidateMasterCard | Validate MasterCard credit card                 |

| Credit Card Issuer | Valid Prefixes | Valid Length |
|--------------------|----------------|--------------|
| VISA               | 4              | 13,16        |
| MasterCard         | 51,52,53,54,55 | 16           |
| AMEX               | 34,37          | 15           |

##### Information:

- Check for the correct prefix and card type
- Apply LUHN formula to ensure it's not a random number.

#### 2. Create a client application

**Objective:** Create a C# client application to invoke the credit card validation process.

3. Create Authentication Process

**Objective:**

- a) Create a form-based authentication system in ASP.NET using Basic Authentication, preferably employing a database lookup to verify valid users.
- b) Validate the user by utilizing the ASP.NET form-based login (logon.asp) from a client application.

4. Create another XML Web service method from the requirements above.

**Objective:** Utilize your creativity to develop additional business functions as outlined in the requirements above.